Phizura: On-the-Fly Music Recording with Method Proxies

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Abstract

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Keywords

Pharo, Method Proxies, Coypu, On-the-fly music programming

1. Introduction

This paper introduces the Phizura Playground, a tool that extends the Pharo Playground with the capability to record the code written and evaluated within it. The recording system is implemented using Method Proxies [1].

The Phizura Playground has been designed to work with Coypu[2], a library and dialect for on-the-fly music programming within the Pharo environment. Coypu enables live coding performances in which musical structures are generated and modified interactively.

Capturing a musical performance without recording the audio output into large files has been a challenge since the early versions of Coypu. A first recording mechanism was implemented by Redwane Engels during an internship, and was demonstrated at ESUG 2024 in Lille.

This initial solution relied on the Pharo Announcer and Announcement classes, which implement the observer design pattern. Despite working reliably, this approach required intrusive modifications to the Coypu codebase and made it harder to extend the system with new behaviors. In recent months, we have reimplemented the recorder using Method Proxies. This redesign removes instrumentation from the system and provides a more flexible framework for extending live coding performances.

2. Objectives

While developing Coypu, we observed that the Pharo language and environment provide the means to easily record the code written and evaluated during a musical performance. This feature is novel in the live coding community and allows long performances to be stored in very small text files. These recordings can then be replayed by other musicians using the same audio server [3] and sound synthesis setup, or with a different *orchestra*[1]. Such lightweight text files enable easy sharing between users and allow concerts to be reproduced even in the absence of the human performer.

We initially implemented a recording mechanism in Coypu based on Announcers and Announcements. However, this solution was intrusive and tightly coupled to the Coypu codebase, making it difficult to extend and maintain, although it functioned as expected.

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[1]We use the term orchestra denotes the set of sound synthesis definitions, instruments, and audio effects used to render a score, following Csound terminology.

To address these limitations, we reimplemented the recorder using Method Proxies. This approach cleanly separates instrumentation from performance-related methods, resulting in a more modular and extensible design.

3. Design and Implementation

To support recording of live evaluations, we developed a custom extension of the Pharo Playground. This was necessary to gain control over expression evaluation without modifying the core environment.

The extended Playground provides a hook on each evaluated expression before execution. This hook captures the evaluated command together with a timestamp. When recording is active, each evaluation is forwarded to a dedicated handler. The handler creates a structured record containing the evaluated expression and its timestamp, representing an absolute point in time using `DateAndTime now`. During replay, temporal spacing is reconstructed by computing the difference between consecutive timestamps, yielding a `Duration` used to reproduce the original timing of the performance.

The handler is also connected to the Playground context, allowing it to access the current editor state. This enables associating each evaluation with the current state of the live coding session.

To control the recording workflow, four icons were added to the Playground toolbar. The first starts a new recording session, the second stops the current session, and the third clears the recorded data. The fourth opens the recorded session in a new Playground, where captured evaluations can be inspected, replayed, or edited as a standalone script.

The resulting design cleanly separates the recording logic from the execution semantics of the Playground. It allows evaluation capture to be added without invasive changes to the language infrastructure.

4. Conclusion

Using Method Proxies to implement a recorder for live coding performances in Coypu required only a small number of classes and methods, and was achieved without modifying the Coypu codebase. The recorder captures every expression evaluated in the Playground, which opens up possibilities beyond music. Any live coding practice conducted within a Pharo Playground can be recorded with the same mechanism, including, for example, graphics rendered with Bloc. The tool can equally serve as a general prototyping recorder, capturing the sequence of evaluations made during an exploratory session in Pharo. The library ships with a test suite and integrates without issues into the existing environment. Future work will focus on exposing a simpler API within the Phizura package, allowing users to extend the instrumentation layer. This will enable the definition of custom actions triggered either by evaluated expressions or by specific methods within a user's own library, giving live coders finer control over what is captured and how it is processed.

References

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